

BIF401 Current Solve Papers

All Important Repeated Questions

Created By: Amir Ahmadani

Q. Define Top down proteomics?

Ans: "Top down proteomics can handle whole proteins".

Top down proteomics measures the intact proteins followed by peptides after fragmentation.

Q. Need for Chau fasman algorithm?

Ans: Chau fasman algorithm is a propensity base method. The method is based on analyses of the relative frequencies of each amino acid in alpha helices, beta sheets, and turns based on known protein structures solved with X-ray crystallography. From these frequencies a set of probability parameter were derived.

Q. Write down an objective base method and clustering approach?

Ans: UPGMA is a clustering approach and least square distance is an objective base method.

Q. Differentiate between finger printing and short gun proteomics?

Ans: Shotgun proteomics digests the entire protein mix 1st followed by peptide analysis & protein database search

- Peptide mass fingerprinting involves protein separation followed by a single protein's peptide analysis.

Q. what are substitutions and indels?

Ans: Indels:

Removal and addition of amino acids in proteins and nucleotides in DNA, RNA by using Gaps named as Indels.

Substitution:

Replacement of a single or a specific pair of amino acid or nucleotide with other on in a sequence is called substitution.

Q. How can pseudoknots be detected in the prediction of RNA?

Ans: Tertiary or 3' structure of RNA may form pseudoknots. To detect the pseudoknots in RNA structure we need "circular plot" which is a graphical approach.

Q. Formation of protein sequence and limitation?

Ans: There are two basic techniques for sequence of a protein,

1. Edman degradation

2. Mass spectrometry

Edman degradation is used to sequence the smaller length protein, about 40 – 50 amino acids.

Large protein sequence is carried out by mass spectrometry.

*Limitations include,

Q. Uniprot and EMBL or EMBL?

Ans> Uniprot is a public data base which is used to find the sequence of protein.

EMBL is genome search engine which is used to search the genome of every recorded species. OR it might be EMBL so,

Input to FASTA search can be in

* EMBL*, = European molecular biology laboratory (EMBL)

Q. Mass spectrometer?

Ans> Mass spectrometer is used to measure mass/charge ratio of ionized proteins and peptides.

Data output from the MS comprises of m/z ratios and intensities of each molecule that is measured.

Function of Tandem MS:

Tandem MS helps in measurement of mass to the fragments as well. This process provides another step in further scoring and ranking and Protein identification thus becomes easier.

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Q. Define Orthology?

Ans> it is a homologous gene present in two organisms that encode protein with same function.

It is evolved by direct vertical descent.

Q. What does role of 5' cap and 3' poly A tail?

Ans: the 5' end of molecule capped with 7 methyl guanosine tri phosphate and 3' end poly A tail cap, both caps play major role to translate mRNA to ribosome, also protect mRNA.

Q.: Explain applications of bioinformatics.

There are some major applications of bioinformatics:

In this field bioinformatics help us in gene finding, in assemblage of genes and forming of databases.

2. Proteomics:

It helps us to decoding protein sequences for better understanding about protein structure, protein to protein relationship, post translational changes and in generating databases for their sequence and structure.

3. Evolutionary study:

It can make phylogenetic trees to find evolutionary relationship between species, also show ancestry between them,

4. System biology:

Bioinformatics also helps us in regulatory mechanism in genes and proteins. So that we can better understand about regulators and treat them by evaluate drugs.

Bioinformatics also help us in further many aspects like,

Drug development

Waste cleanup

Gene therapy

Preventative medicine etc.

Q=write types of phylogenetic trees ? no option

Rooted trees:

- ☐ Each node with descendants represents the inferred most recent common ancestor of the descendants.
- ☐ The edge lengths in some trees may be interpreted as time estimates.
- ☐ Rooted trees can show temporal evolutionary direction.
- ☐ Expensive.

Unrooted:

- ☐ Only the relatedness of the leaf nodes.
- ☐ Do not require the ancestral root to be known or inferred.
- ☐ Less expensive.

Uses:

In bioinformatics, such as

- ☐ Rooted and unrooted trees can be used to show phylogenetic relationships between sequences.

Q. define open reading frame?

In molecular genetics, an open reading frame (ORF) is the part of a reading frame that has the potential

to code for a protein or peptide. An ORF is a continuous stretch of codons that do not contain a stop

codon (usually UAA, UAG or UGA).

Q=what is FASTA? Why we use it?

- FASTA can perform quick comparison of protein and nucleotide sequence
- Can also perform genome and proteome similarities search
- Its also available online like BLAST.

FASTA can search sequence databases and identify unknown sequences by comparing them to the

known sequence databases. This can help obtain information on the parent organism, function and

evolutionary history

Q.What is BLAST ? NO OPTION

➤ BLAST finds regions of similarity between biological sequences. The program compares nucleotide or protein sequences to sequence databases and calculates the statistical significance. In bioinformatics, BLAST for Basic Local Alignment Search Tool is an algorithm for comparing primary biological sequence information, such as the amino acid sequences of proteins or the nucleotides of DNA sequences. A BLAST search enables a researcher to compare a query sequence with a library or database of sequences, and identify library sequences that resemble the query sequence above a certain threshold.

➤ Different types of BLASTs are available according to the query sequences. For example, following the discovery of a previously unknown gene in the mouse, a scientist will typically perform a BLAST search of the human genome to see if humans carry a similar gene; BLAST will identify sequences in the human genome that resemble the mouse gene based on similarity of sequence.

Q. What is the difference between basic and acidic amino acids?

Ans: the difference is in their side chains, the acidic ones have acidic side chain while basic ones have basic side chain at neutral pH.

- Arginine, lysine and histidine are the basic ones.
- Aspartate and glutamate are the acidic ones.

Q. What is scope of bioinformatics?

Ans. Bioinformatics primarily deals with digitalized biological information as well as data reported from biology experiments. Computational methods, data processing techniques and algorithms are used in addressing the following issues,

Use of computational algorithms and techniques for:

1. Storage,
2. Organization,
3. Analysis, and
4. Representation of biological information

How junctions are formed?

Ans: junctions are formed when two or more double stranded regions of RNA converging to form a closed structure.

Q. Name of famous protein data base?

Ans: SWISS PROT and UNI PROT

Q. Four objectives of comparing sequence?

Ans: 1. Similarities among sequences

2. Differences b/w sequences

3. Evolutionary history

4. Predict the function of molecules

Q. How bulges are formed?

Ans: in RNA 2' structure bulges are formed when double stranded region cannot form base pairs perfectly.

Q. how biological simulation help community?

Ans: it helps community by predicting disease, progression and outcomes.

Q. what is role of EXPASY?

Ans: it provides access to a variety of online data bases and tools depending upon our requirements. We can find sequence information by it.

Q. Define PSTs? No option

Ans: peptide sequence tags (PSTs) are actually small amino acid sequence of peptides producing during MS2.

Q. What is meant by "PK" value of an amino acid?

Ans: PK is the value for an amino acid is that PH at which exactly half of amino acids are charged and half are not charged.

Q. how loops are found after the finding of alpha and beta sheets?

Ans: After computing the propensity of alpha helices and beta sheets, we need to settle for loops.

□ Let's see how can we find out the loops using Chou Fasman Algorithm.

- For any j th residue in sequence, we calculate $f(\text{Total}) = f(j) f(j+1) f(j+2) f(j+3)$ (tetra peptide).
- If,
- $f(\text{Total}) > 0.000075$.
- The average value for $P(\text{turn}) > 1.00$ in the tetra peptide.
- The averages for the tetra peptide are such $P(\text{a-helix}) < P(\text{turn}) > P(\text{b-sheet})$,
- It is a Turn!

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Q. . How protein structure identified?

Ans: If another protein which has a similar sequence also has its structure known, the

structure of an unknown protein can be predicted based on that similar protein.

So, it is then possible to identify unknown protein structures by just examining the homologous protein sequences. Good sequence alignment and identity ensures that homology modelling will give accurate results

Thus, Homology modeling is used to predict structures of proteins having high sequence similarity with other proteins with known structures:

Protein structure prediction:

There are three different strategies for structure prediction

1. Homology Modelling
2. Threading/Fold Recognition
3. Ab Initio Modelling.

Q. What are the properties and role of amino acids?

Ans: Properties:

Amino acids have several properties such as charge state, polarity and hydrophobicity. Amino acids not only have physical and chemical properties, but also structural properties.

Role:

These structural properties are equally important in giving rise to protein structures. Since some amino acids are hydrophobic, they may be employed in forming a stable core in a protein.

- Also, chemically inactive amino acids reduce chances of destabilizing reactions in core.

Q. Why folding is important?

Ans: Proteins fold spontaneously. Proteins fold to achieve thermodynamic stability. Proteins fold to

organize themselves for performing functions in cells.

Q. write down the terms of PDB file format.

Ans: HEADER: contain the brief description of the structure, the date and PDB ID code.

TITLE: title of the structure.

SOURCE: identifies which organism the structure came from.

COMPND: brief detail of the structure.

REVDAT: the date of the last revision.

JRNL: one or more literature reference that describe the structure.

KEYWDS: list the set of useful words/phrases that describe the structure.

AUTHOR: the scientist depositing the structure.

REMARKS: detail of the experimental method used to determine the structure are contain in this subsection.

Q. define ionization in terms of proteins?

Ans: ionization of protein means to ionize a protein molecule to make it charged by removing an electron from it. Protein ionization is the first step in MS-based proteomics protocols

- Ionization process involves loading of a proton onto a protein or the removal of a proton from the protein
- Ionization results in an increase or decrease of the protein/peptide mass

Two primary techniques used for ionization of protein,

1. Electrospray ionization (ESI)
2. Matrix - assisted laser desorption/ionization (MALDI)

Q. Briefly describe Chou Fasman algorithm?

Ans: Chou – Fasman method:

It is a technique for the prediction of secondary structures in proteins i.e.

Alpha Helices, Beta Sheets and Turns is Chou – Fasman technique. The method is based on analyses of the relative frequencies of each amino acid in alpha helices, beta sheets, and turns based on known protein structures solved with X-ray crystallography. From these frequencies a set of probability parameters (in our handouts, it is propensity table) were derived

☐ For the appearance of each amino acid in each secondary structure type.

☐ To predict the probability that a given sequence of amino acids would form a helix, a beta strand, or a turn in a protein.

Q. How can alpha helix be expressed?

Ans: Chou-fasman algorithm (alpha helix):

- ☐ For Alpha Helices, 4 contiguous amino acids are required.
- ☐ Their Alpha-Helix propensity should be more than 1.0
- ☐ Once this propensity falls below 1.0, Alpha-Helix stops.

Q. Describe scores in silico fragmentation?

Ans: Silico fragmentation scoring:

- ☐ Count the matches between in silico and in vitro peaks.
- ☐ Give an equivalent score to the candidate protein.
- ☐ Weigh each of the aforementioned match by the mass error.
- ☐ Accumulate the score

Q. what is 3D_1D bowie algorithm?

Ans: Proposed by Bowie et al in 1991

- Converts 3D structure into a 1-D string profile for each structure in the fold library
- Align the target sequence to these profiles 3D-1D methods convert structure and environment information into "profiles"
- Score for each amino acid is computed for each profile

Define Peptide mass finger printing nd shotgun?

Ans: Shotgun proteomics digests the entire protein mix 1st followed by peptide analysis & protein database search

- Peptide mass fingerprinting involves protein separation followed by a single protein's peptide analysis.

Write steps of protein sequence identification? OR Enlist protein sequence

methodAns: Mass spectrometers are used to measure the molecular weight of proteins and peptides.

Following steps involve in protein sequence identification,

1. complex protein
2. separation
3. ionization by mass spectrometry
4. fragmentation MS2
5. mass spectra
6. EST Determination
7. filter protein database
8. in silico fragmentation of candidate protein

9. matching of experimental and insilco peak list

10. post translational modification

11. protein score.

Q. write types of RNA ?

ans: Coding RNAs

- Coding RNAs as is obvious from their name , code for proetein
- Non-coding RNAs
- Non-coding RNAs regulate/assist in the process of translation.Q

Other types of RNA are,

1= m RNA

2=t RNA

3= r RNA

4= miRNA

5= si RNA

Advantage and disadvantage of Ab initio:

Advantages:

- ☐ Ab Initio methods can fold any target sequence using only physical atomic properties.
- ☐ Predictions are mostly accurate and correctly describe the natural folding process. Disadvantages:

- ☐ Ab initio methods are the very difficult to design (energy function).
- ☐ These methods are slow due to the huge possibilities

PAM matrix

- ☐ PAM means "Point Accepted Mutations"
- ☐ Point accepted mutation is a substitution of one amino acid by another such that the protein functions stays conserved.
- ☐ PAM unit is a time in which about 1% of amino acids in a sequence undergo accepted mutations

□ PAM matrices are scoring matrices that are useful in computing sequence alignment scores.

STEP TO COMPUTE PAM MATRICES

1. Align the protein sequence which are 1-PAM Unit diverge.
2. Let $A_{i,j}$ be the number of times A_i is substituted by A_j .
3. Compute the frequency f_i of amino acid A_i .

Then, $PAM1 = P_{ij} = \frac{A_{ij}}{\sum_k A_{ik}}$

$$\sum_k A_{ik}$$

PAM 'n' = (PAM1)

Steps present in flow chart of homology modeling: - (10 marks)

1. Template recognition and initial alignment.
2. Alignment correction.
3. Backbone generation.
4. Loop modeling.
5. Side-chain modeling.
6. Model optimization.
7. Model validation

How homology modeling is used for knowing the sequence of unknown protein sequence: (10 marks)

If another protein which has a similar sequence also has its structure known, the structure of an unknown protein can be predicted based on that similar protein.

So, it is then possible to identify unknown protein structures by just examining the homologous protein sequences. Good sequence alignment and identity

ensures that homology modelling will give accurate results

Thus, Homology modeling is used to predict structures of proteins having high sequence similarity with other proteins with known structures:

How we can get scoring in BUP proteomic

There are two approaches to perform BUP:

- ☐ Peptide Mass Fingerprinting.
- ☐ Shotgun Proteomics

Score in Protein Fragmentation:-

If we can:

- ☐ Measure the mass of fragments using MS.
- ☐ Calculate the Protein Fragmentation Techniques theoretical mass of the fragments.

Then, we can award score on the basis of the similarity of experimental and theoretical mass

Protein Fragmentation techniques:

- ☐ Electron Capture Dissociation (ECD).
- ☐ Electron Transfer Dissociation (ETD).
- ☐ Collision Induced Dissociated (CID).

Types of secondary structures of RNA:

1. Single stranded:

- ☐ 3' end may fold on to the 5' end.

2. Helices:

- ☐ Double stranded RNA helix of stacked base pairs

3. Hairpin loop:

- ☐ The loop of the hairpin must at least four bases long to avoid steric hindrance with base-pairing in the stem part of the structure

4. Bulge Loops:

- ☐ Bulges, are formed when a double-stranded region cannot form base pairs perfectly

5. Interior loop:

- ☐ Interior loops are formed by an asymmetric number of unpaired bases on each side of the loop.

6. Junctions or intersections:

- ☐ Junctions include two or more double-stranded regions converging to

form a closed structure.

□ The unpaired bases appear as a bulge.

Q. How to build scoring matrices?

Ans: To build the Scoring Matrices we analyze the amino acids and nucleotides which are substituted in single gene and protein sequence.

Scoring Matrices have both values +ve and -ve. Positive value for matches and negative value for mismatches. Different type of scoring matrices can be developed based on underlying strategy.

1. Each amino acid have different property.
2. Each amino acid have different frequency.
3. When we compare the sequences they match and mismatch according to their frequency.

Q. Optimal energy function in structural prediction?

Ans: Energy based methods involve evaluating the free energy structures. To compute the RNA sequence for 1' or 2' optimal structure prediction we use Zuker's Algorithm.

RNA Secondary Structure Prediction.

Zuker's Algorithm helps us to compute the stabilizing energies (-ve) and also destabilizing energies (+ve values). And also compute the sum of +ve and -ve energies. All possible 2' structures are generated. The best 2' structure is selected.

Q. Home message of this course?

- Ans: 1. Introduce the classical algorithms in bioinformatics
2. Link them to latest developments in the field
 3. Evaluate the future applications

Q. Diff. b/w MS and Raw file data?

Ans: Mass spectrometer (MS) outputs data with mass/charge ratios & respective ion intensities. RAW file is a format in which an instrument outputs data in binary form. Several software exist for converting RAW file formats into open software formats. Each open format has its own unique advantages. mzXML and MGF formats are most frequently used

Q. Why we need make protein structure when we have X-Ray and NMR?

Ans: There are thousand of sequences and lesser structures are available. Reason is that some proteins cannot be crystalized or it may not be soluble so we have need to make protein structure althou we have x- ray crystallography.

Q. Amino acid with 3 codon?

Ans: lysine has 3 codon. (Lie = AUU, AUG, AUA)

Q. How + and – energies released in molecule?

Ans: The stabilizing energy associated with stacking base pairs in a double-stranded region is (-ve) energy.

- The destabilizing influence of unpaired regions is (+ve) energy.

Q. How back bone of protein form?

Ans: C termini in a protein backbone C-Alphas can be used to construct the backbone of a protein towards its visualization. Proteins have Carbon and Nitrogen in their backbone

Q. which factor consider during N-J trace back?

Ans: 'Trace back' strategy is used to recover the optimal structure, there can be multiple trace backs. Each trace back can be used to construct an RNA secondary structure. NJ Select the trace back with the highest number of coupled nucleotides.

Q. Top down proteomics?

Ans: for a direct measurement of protein mass

- Solution: TDP

1. Samples containing the protein mixture from cells or tissue are obtained

2. The entire protein mix is analyzed for protein masses (MS1)

3. The mass list thus obtained has masses of all intact proteins4. Note that proteins typically carry post-translational modifications and TDP caters for their mass measurement as well

5. After MS1, one protein is selected at a time and fragmented to obtain its peptides (MS2)
6. Several protein fragmentation techniques exist for fragmenting proteins
7. Fragmentation products i.e. peptides are measured for their mass (MS2)
8. The process is repeated (MS'n')

Conclusions:

- TDP measures the mass of intact proteins
- Mass of post translational modifications can be easily measured.

Q. Formula of scoring matrices?

Ans: $p_{ij} = A_{ij}$

$\sum_k k A_{ik}$

Q. What are forces used in amino acids to fold a protein?

Ans: Electrostatic interactions

- van der Waals interactions
- Hydrogen bonds
- Hydrophobic interactions

Q. How OH of RNA make it less stable than DNA?

Ans: The two hydroxyl groups make RNA less stable than DNA because it is more prone to hydrolysis Hence, RNA has a short life span and is more prone to degradation.

REMEMBER ME IN YOUR PRAYERS

BIF401 - Bioinformatics I.

1.What Are Uses Of Mass Spectrometry? 2

Mass spectrometry (MS) is an analytical technique that ionizes chemical species and sorts the ions

based on their mass-to-charge ratio. In simpler terms, a mass spectrum measures the masses within a

sample. Mass spectrometry is used in many different fields and is applied to pure samples as well as

complex mixtures.

2. Need For Chou Fasman Algorithms? 2

The Chou-Fasman algorithm is simple in principle. The conformational parameters for each amino acid

were calculated by considering the relative frequency of a given amino acid within a protein, its occurrence in a given type of secondary structure, and the fraction of residues occurring in that type of

structure.

3. Three individual scores Integrative Scoring Scheme? 2

Three individual scores can be obtained including

1. MW Match Score

2. PST Match Score

3. In vitro & In silico Match Score.

4. Write benefits of phylogenetic studies marks • 03

5. Advantage of Folded Protein over Denatured Protein. 5

Difference Between Folding Protein and Denature Protein, / What Factors Are Used For Folding Proceeds?

Protein folding is the physical process by which a protein chain acquires its native 3-dimensional structure, a conformation that is usually biologically functional, in an expeditious and reproducible manner.

Denaturation of proteins involves the disruption and possible destruction of both the secondary and

tertiary structures. Since denaturation reactions are not strong enough to break the peptide bonds, the

primary structure (sequence of amino acids) remains the same after a denaturation process.

Protein folding is a process by which a polypeptide chain folds to become a biologically active protein in

its native 3D structure. Protein structure is crucial to its function. Folded proteins are held together by

various molecular interactions. ... The amino acid sequence of a protein determines its 3D structure.

The forces involved in protein folding include: • Electrostatic interactions, • van der Waals interactions, • Hydrogen bonds, • Hydrophobic interactions

6. Score in Silico fragment scoring? .5

- Count the matches between in silico and in vitro peaks.
- Give an equivalent score to the candidate protein.
- Weigh each of the aforementioned match by the mass error.
- Accumulate the score

7. How Loops Are Found After The Finding Of Alpha And Beta Sheets? .5

Alpha helices and beta strands are connected by these turns and loops. ... Loops and turns generally

lie on the surfaces of proteins so they often participate in interactions between proteins and other molecules. In a loop, there are no regular structures as can be found in helices or beta strands.

8. Tandem mass spectrometry

Tandem mass spectrometry (MS-MS) is a related technology in which compounds are separated by

molecular weight by one mass spectrometer, fragmented as they exit, and identified on the basis of

their fragments by a second mass spectrometer

9. Advantage and disadvantage of Ab initio method?

Advantages:

Ab Initio methods can fold any target sequence using only physical atomic properties.

Predictions are

mostly accurate and correctly describe the natural folding process.

Disadvantages:

Ab initio methods are the very difficult to design (energy function). These methods are slow due to

the huge possibilities.

10. Types of secondary structures of RNA:

1. Single stranded:

3' end may fold on to the 5' end.

2. Helices:

Double stranded RNA helix of stacked base pairs

3. Hairpin loop:

The loop of the hairpin must at least four bases long to avoid steric hindrance with base-pairing in the

stem part of the structure

4. Bulge Loops:

Bulges, are formed when a double-stranded region cannot form base pairs perfectly.

5. Interior loop:

Interior loops are formed by an asymmetric number of unpaired bases on each side of the loop.

6. Junctions or intersections:

Junctions include two or more double-stranded regions converging to form a closed structure. The

unpaired bases appear as a bulge

11. How homology modeling is used for knowing the sequence of unknown protein sequence?

If another protein which has a similar sequence also has its structure known, the structure of an unknown protein can be predicted based on that similar protein. So, it is then possible to identify unknown protein structures by just examining the homologous protein sequences. Good sequence

alignment and identity ensures that homology modelling will give accurate results. Thus, Homology

modeling is used to predict structures of proteins having high sequence similarity with other proteins

with known structures:

12. Role of Amino Acid?

Protein plays a crucial role in almost all biological processes and amino acids are the building blocks of it.

A large proportion of our cells, muscles and tissue is made up of amino acids, meaning they carry out

many important bodily functions, such as giving cells their structure.

13. Write Down The Following Terms Of PDB (3)

Title (Title of the structure)

Source (Identify which organism structure from)

COMPND (Brief detail of the structure)

REVDAT (The data of last revision)

14. What Happend When Ph Is Less Than The Pk Value?

If the pH is higher than the pKa, then the compound will be deprotonated. A further consideration is the

charge on the compound. Acids are neutral when protonated and negatively charged (ionized) when

deprotonated. Bases are neutral when deprotonated and positively charged (ionized) when protonated.

15. Chou Fasman Algorithm?

The Chou–Fasman method is an empirical technique for the prediction of secondary structures in proteins,

originally developed in the 1970s by Peter Y. Chou and Gerald D. Fasman.

16. Function of Tandem MS?

Tandem MS helps in measurement of mass to the fragments as well. This process provides another step in

further scoring and ranking and Protein identification thus becomes easier.

17. Factors That Participate In Folding Protein?

The external factors involved in protein denaturation or disruption of the native state include temperature,

external fields (electric, magnetic), molecular crowding, and even the limitation of space, which can have a

big influence on the folding of proteins

18. Need For Chou Fasman Algorithms?

The Chou-Fasman algorithm is simple in principle. The conformational parameters for each amino acid

were calculated by considering the relative frequency of a given amino acid within a protein, its occurrence

in a given type of secondary structure, and the fraction of residues occurring in that type of structure.

19. Optimal Energy Function In Protein?

Results: For a single protein sequence, the probability of success (i.e. the probability that the folded state

is the lowest energy state) is derived. We then maximize the average probability of success for a set of

proteins to obtain the optimal potential energy function

20. Silico Fragments Comparison?

Matching experimental fragments with in silico fragments is the final resort in protein search & identification

21. Chou – Fasman method?

It is a technique for the prediction of secondary structures in proteins i.e. Alpha Helices, Beta Sheets and

Turns is Chou – Fasman technique. The method is based on analyses of the relative frequencies of each

amino acid in alpha helices, beta sheets, and turns based on known protein structures solved with X-ray

crystallography. From these frequencies a set of probability parameters (in our handouts, it is propensity

table) were derived

For the appearance of each amino acid in each secondary structure type. To predict the probability that

a given sequence of amino acids would form a helix, a beta strand, or a turn in a protein.

Chou-fasman algorithm (alpha helix):

For Alpha Helices, 4 contiguous amino acids are required.

Their Alpha-Helix propensity should be more than 1.0

Once this propensity falls below 1.0, Alpha-Helix stops.

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22. Improvements Of Chou Fasman Algorithm?

The Chou-Fasman algorithm, one of the earliest methods, has been successfully applied to the prediction.

... We improved Chou-Fasman method in three aspects. (a) Replace the nucleation regions with extreme

values of coefficients calculated by the continuous wavelet transform.

23. What is hydrophobicity of amino acid?

In chemistry, hydrophobicity is the physical property of a molecule that is seemingly repelled from a mass

of water. In contrast, hydrophiles are attracted to water. Hydrophobic molecules tend to be nonpolar and,

thus, prefer other neutral molecules and nonpolar solvents.

24. how propensity table can helps us?

25. Seven Steps present in flow chart of homology modeling?

10

1. Template recognition and initial alignment.

2. Alignment correction.

3. Backbone generation.

4. Loop modeling.

5. Side-chain modeling.

6. Model optimization.

7. Model validation

26. Role of Amino Acid?

Protein plays a crucial role in almost all biological processes and amino acids are the building blocks of it.

A large proportion of our cells, muscles and tissue is made up of amino acids, meaning they carry out

many important bodily functions, such as giving cells their structure.

27. Role of Global Protein?

Global protein function prediction from protein-protein interaction networks. ... The availability of entire

genome sequences and of high-throughput capabilities to determine gene coexpression patterns has

shifted the research focus from the study of single proteins or small complexes to that of the entire

proteome.

28. Acidic and Basic Amino Acid?

There are three amino acids that have basic side chains at neutral pH. These are arginine (Arg), lysine

(Lys), and histidine (His). Their side chains contain nitrogen and resemble ammonia, which is a base. Their

pKa's are high enough that they tend to bind protons, gaining a positive charge in the process.

29. Points of Bottom up Proteomic?

The major alternative workflow used in high-throughput proteomics is called top-down proteomics and does

not use proteolytic digestion. Essentially, bottom-up proteomics is a means of determining the protein

make-up of a given sample of cells, tissues, etc. (Internet)

30. Varies RNA Secondary Structure?

Types of RNA Secondary Structures - I

- RNA Complementary bases in primary structure bond together to fold RNA into a secondary structure

Types of RNA Secondary Structures – II

- RNA 1' sequence (5' – 3') can fold onto itself to make 2' structures.

- Two previous examples of 2' structures include Helix and Hairpin structures.

Types of RNA Secondary Structures - III

- RNA can be fold to form helices, bulge loops, and interior loops. An other 2' RNA structure is the Junction

or Intersection.

31. Objective Base Method and Clustering Approach?

Clustering Approach UPGMA WPGMA Neighbor Joining Single Linkage Complete Linkage Objective

based Methods Least Square Distances Maximum Likelihood Maximum Parsimony

32. Shot Gun and Peptide Mass Finger Printing?

Shotgun proteomics digests the entire protein mix 1st followed by peptide analysis & protein database

search Peptide mass fingerprinting involves protein separation followed by a single protein's peptide

analysis.

33. substitution and Indels?

Substitution at adjacent nucleotides (primarily substitutions at two adjacent nucleotides, but substitutions at

three adjacent nucleotides have been observed.

Indels, being either insertions, or deletions, can be used as genetic markers in natural populations,

especially in phylogenetic studies.

34. Prediction of Protein Structure?

Protein structure prediction is the inference of the three-dimensional structure of a protein from its amino

acid sequence—that is, the prediction of its folding and its secondary and tertiary structure from its primary

structure.

35. How Can We Predict Amino Acid In Alpha Helix And Beta Sheets?

Proteins that convert from alpha helix to beta sheet: implications for folding and disease. The sequence of

a protein normally determines which amino acid residues will form alpha helices, and which one beta

sheets, to an extent that allows secondary structure prediction to be made with a reasonable reliability..

36. Rooted Un Rooted Tree?

Rooted Tree:

Each node with descendants represents the inferred most recent common ancestor of the descendants.

- The edge lengths in some trees may be interpreted as time estimates.
- Rooted trees can show temporal evolutionary direction.
- Expensive.

Unrooted:

- Only the relatedness of the leaf nodes.
- Do not require the ancestral root to be known or inferred.
- Less expensive.

Uses: In bioinformatics, such as

Rooted and unrooted trees can be used to show phylogenetic relationships between sequences..

37. Molecular biology?

It is the branch of biology that deals with the structure and function of the macromolecules (e.g. proteins

and nucleic acids) essential to life. Molecular biology is the study of biology at a molecular level.

38. Protein Fragmentation techniques?

- Electron Capture Dissociation (ECD).
- Electron Transfer Dissociation (ETD).
- Collision Induced Dissociated (CID).

39. Protein structure prediction?

There are three different strategies for structure prediction

1. Homology Modelling
2. Threading/Fold Recognition

3. Ab Initio Modelling

BIF401 Current Solve Papers

Created By: Amir Ahmadani

IMPORTANT MCQS OF BIF401

MCQS

- 1) Fasta Was Develop In(Ans1988)
- 2) In Corey Paulin Modal Of The Protein Each Atom Is Represented By(Solid Sphere)
- 3) How Many Storages Use For Structural Modling (3)
- 4) Modeler Software Use For Homology Modling Which Use(Phython File Format,Not Confrm)
- 5) Mass Of The Based Proteomic Begin With Measure Of(Intact Protein Amino Acid)
- 6) Protein Sequence Obtained From(Uni Port)
- 7) During The Formation Of The 5 Nucleotide Hydrogen Bond Energy Release(-12kcal/Mol)
- 8) Which Method Is Not Use For Making Polygenatic???Msa
- 9) Rna Dont Have Structer???Quatnaryy
- 10) Large Numbr Of Sequence Availble??Publically
- 11) Coding Region Of Dna Is???Exon
- 12) Blast Use For???Unknown And Known Protein
- 13) Expasy Develop By???Sib
- 14) Variation In Sequence Develop By??? Insertion,Deletion And Substituin
- 15) Bond Form Between C And N Terminas??? Hydrogen Bond
- 16) X.Y.Z Position Of Protein Are Available By??? Pdb
- 17) Technique Use To Measure The Mass Of Protein??? Mass Stractrometry

- 18) Large Amino Acid??? Tryptophan
- 19) Positive Charge Amino Acid??? Lysine And Arginine
- 20) Mrna Has??? Single Strand
- 21) C.Alpha Atom Are Traced To Recreat??? 3d Structer Of Protein
- 22) Bond Form Between Two Atom??? Energy Release
- 23) Provide Clue Of Precursor Protein Peptide Sequence??Pst
- 24) Amino Acid Requird For Betasheet??? 3-5
- 25)Are Formed By Bonding Between C_i And N_{i+4n} Protein Backbone??? Alpha Halix
- 26) Blasta Stand For??? Basic Local Alignment Search Tool
- 27) Amino Acid Have Uniwe Trait Which Can Help In??? Protein Folding
- 28) Align Comparison Between dna/Dna And dna/Protein??? Fasta.
- 29) Ratio Use To Calculate Mass Of Protein??? M/Q.
- 30) Treading Not Requied For??? Structer Of Entire Backbone.
- 31) Needle Man Wounch Man Or Smith Water Man Algorithm Use For???
Local And Global Alignmnt.
- 32) Dianamic Programing Recombinant Nucleotide By??? Traseback.
- 33) Insertion And Deletion Cause??? Gap.
- 34) Information Required For Follding Of Amino Acid Is Present In??? Amino Acid.
- 35) Mgf Use For??? Proteomic.
- 36) Nj Algorith Use To Pridict The??? 2nd Structer Of Rna..
- 37) Genomics, evolutionary studies and system biology are application of bioinformatics
- 38) Exact matching require which type of nucleotide-same number of nucleotide
- 39) Accurate solution of protein structure is one of the ____ toughest problems
- 40) Tract back is overlap matches can start from any position in scoring matrix
- 41) The matrix hasscores. Positive and negative
- 42) MASCOT ____ can search people mass finger printing and shotgun proteomics data set
- 43) Spectrometer measures the protein by _____ Mass/charge
- 44) Simulation of the folding process depends _____ Energy function

- 45)Information about DNA/RNA is available on gene bank
- 46)TDP is used to measure the molecules weight of intact proteins.
- 47)Coordinates of alpha carbon in the protein backbone can be used for structural visualization.
- 48)in structure visualization c-alpha atoms are traced to recreate a _____3D protein structure
- 49)in UPGMA distance is calculated btw _____ two clusters and between two trees
- 50)pk value of Aspartic acid 3.9
- 51)Largest amino acid Tryptophan
- 52)2
- nd STRUCTURE of proteins can be obtained by using_____ Chou Fasman algorithm
- 53)NJ Algorithm strategy is used to predict __ RNA 2` structure
- 54)The loop of hair pin must be at least ____ in length 2 aminoacids
- 55)Fold recognition is also calledthreading
- 56)Ab initio method in contrast ,base their prediction on low energy model.
- 57)Bioinformatics require.....smart mind, and connected to internet, both,
- 58)Sequence alignment tool is.....PROSIGHT and MASCOT
- 59)MS1,MS2 provide us data identifying unknown.....Proteins sequence.
- 60)MGF files develop as an open standard for..... proteomics data.
- 61)PDB coordinates alpha carbon in protein back bone can be used for comparison .
- 62)Alpha carbon atom can be obtained from.....PDB
- 63)in formation of 2nd structure of protein C & N can make.....Hydrogen bonds
- 64)information for protein folding is into its native structure is in..... protein's amino acid sequence!
- 65)Energy released during bond formation.
- 66)hydrogen bonding occur in ...secondary ..Protein structure
- 67)Bond between C and H in alpha helices____Hydrogen bonding
- 68)Amino acid having 3 codons_____ isoleucine
- 69)MALDI typically adds a to protein or peptide. Proton
- 70)low quality match gets_____lower score

- 71)_____ are sequences of amino acids produced during MS2. PST (peptide sequence tags)
- 72) MS2 data can be used to extract..... peptide sequence tags.
- 73)The alpha helices propensity should be more than____(1.0)
- 74)How many types are of protein sequencing techniques__ (2)
- 75)Blast can find sequences of _____nucleotides amino acid
- 76)If a protein sequence of 26 amino acids is fragmented at 11 amino acid C11, Z25
- 77)To find out unknown sequence of nucleotide we use . NGS or Mass spectrometry
- 78)How many types of peptide May be injected in mass spectrometry chamber.. . three hundred thousand to four hundred thousand
- 79).MASCOT.can search peptide mass finger printing and shotgun proteomics dataset.
- 80)Stabilizing and destabilizing energy give us quality of 2nd structure .
- 81)The Fasman algorithm is based on statistical occurrence of Amino Acids in known structures.
- 82)DP recombinant the nucleotide recombination through process of Traceback
- 83) MALDI.....add proton to a protein and a peptide..
- 84)Positively charged amino acids are..... 3
- 85)Aromatic amino acid include phenylalanine, tyrosine tryptophan
- 86)Homology modelling fail to predict quality structure
- 87)How many forces are involved in protein folding.
- 88)Pairwise alignments tells the similarity between sequence... by maximizing the matches.
- 89)Dot plots employ dot matrix with two sequence plotted represents.....pairwise alignment and comparison.
- 90)Differentiate b/w DNA and RNA sequence..... RNA has more variety of sequence
- 91)Cell molecule are produced after transformation of DNA to protein
- 92) Dot plot cannot deal with..... Insertion, deletion and gaps.
- 93)Exons are may be more conserved.
- 94)In dynamic algorithm we can do comparison of more sequence. Three
- 95)Mass/Charge ratio helps calculate the mass of the Protein
- 96)In scale tree branches lengths are equal to the magnitude of change in the nodes.

- 97) How many types of peptides mix in MS chamber..... 300,000 – 400,000
- 98) mRNA is a structure. Planer
- 99) When unpaired bases of 2' structure join to form..... 3' structure
- 100) N Jackobson use to pridict 2' structure.
- 101) Aromatic amino acid Tyrosine
- 102) Positive charge amino acids..... Lysine, arginine, histidin
- 103) MS begins with the measure of intact protein
- 104) Amount of amino acids in alpha helix.....(4)
- 105) Intact mass of protein can be found by..... MS1
- 106) MSA can be done by..... CLUSTAL
- 107) Which is used as sequence alignment tool..... PROSIT
- 108) MS1 and MS2 help to identify (PST)
- 109) MGF is used for proteomics
- 110) X,Y,Z positions of alpha carbon available online on (PDB)
- 111) Method to determine RNA structure NMR, X-Ray crystallography
- 112) CLUSTLA runs..... Slow accurate/ fast appropriate
- 113) Which give info about precursor protein..... PSTS
- 114) Raw data files format can be converted to Open format
- 115) Which of the following is cartoonic figure..... Ribbon diagram
- 116) In CATH protein recognize according to their..... structural similarity
- 117) Method for obtaining 1' structure..... (Edman Degradation Tandem Mass Spectrometry).
1. FASTA achieves alignment by using short lengths of exact matches.
 2. Dot plots cannot capture insertions, deletions and gap indications
 3. Dot plots employ dot matrix with two sequences plotted on top and left of the matrix
 4. In multiple sequencing, we have both
 5. CLUSTALW can run in two modes slow/accurate, fast/appropriate.

6. The correct order of progressive alignment for multiple sequencing alignment is Pair wise alignment, Guide tree, branching order in tree.
7. Blast can find sequencing from protein or nucleotide database it can find known and unknown sequencing.
8. : Next generation sequencing is used to known about unknown sequences of nucleotide.
9. In multiple sequence algorithm doing so many progressive alignment can be slow.
10. : Structural information of a molecule obtained by X ray crystallography & Nuclear magnetic resonance.
11. Before starting to mach nucleotide or amino acid are need to find the Location of conserved residues.
12. Bioinformatics is an interdisciplinary relative new development of biology, mathematics, computer science, chemistry and physics.
13. Fasta online software Can also perform genome and proteome similarity search .true
14. From Fasta we can obtain the information on the parent organism, function and evolution history.
15. name any two frequently used scoring matrix Blosum and pam.
16. Amino acids are chained together by peptide bonds
17. To find out the unknown sequence of nucleotide and protein we use blast.
18. GenBank and Uniprot, have A large number of sequences are available in publically .
19. Protein databank is a data base
20. In dynamic programming we can do multiple sequence alignment upto 21 sequence .
21. How many types of secondary structure can be formed possible of RNA? 5.
22. RNA structures have the lowest (or close) quantity of free energy.

23. RNA can play role due to its structure . all of above.
24. Rna contains ribose sugar?
25. Thymine is replaced by uracil in rna.
26. RNA molecules readily degrade due to their short shelf life .
27. Each Nucleotide in rna has covalently bonded to the backbone .
28. Gibbs Free Energy" is the free energy of an RNA molecule available for reaction .
29. The structure of rna sampled can be measured by.Atomic Force Microscopy
30. The 3'-end of mRNA has a poly A tail
31. The role of mRNA Carry genetic information from DNA to Ribosomes where proteins are being assembled .
32. Messenger rna is only 5-10% of rna content in the cell.

BIF401 Current Solve Papers

Created By: Amir Ahmadani

From Mid Term

- 1: Gaps are produced by insertion or deletion.
- 2: Example of energy base Zuker algorithm.
- 3: Two forms of DNA rending sequence.
- 4: Translation involves coding of protein by RNA at ribosome.
- 5: some medicine have side effect on certain patient Frontiers in personalized medicine.
- 6: By comparing sequence we can get similarity, specific difference, relationship, evolutionary insight.
- 7: Non clustering method include Maximum parsimony.
- 8: Destabilizing energy of a nucleotide has (+) value.
- 9: Comparison of protein sequence can be done with...Uniprot
- 10: Function is a property of the RNA structure,which is formed from the RNA sequences.

- 11: Structural information of a molecule obtained by resonance.
- 12: Protein have 20 amino acid.
- 13: If sequence are 9 then according formula it will be 9^2 .
- 14: Fasta is available online.
- 15 Fasta was developed in 1988.
- 16: Bioinformatics is the inter disciplinary relative new developing field.
- 17: Genbank can be reached by...online website or www.ncbi.nlm.nih.gov/genbank.
- 18: Before starting to match nucleotide or amino acid are need to find the Location of conserved residues.
- 19: some medicine have side effect on certain patient Frontiers in personalized medicine.
- 20: Identity is the number of nucleotide or amino acid which match exactly between two biological sequence.
- 21: Alignment are represented by diagonals in dot matrix.
- 22: Needleman Wunsch algorithm we use for..... Global alignment.
- 23: Exons in protein tend to be more conserved then the intron.
- 24: In MSA the distance between 2 sequences is equal to NO of identical residue/number of aligned residue.
- 25: All the changes which accrued in our cell metabolism
- 26: The loop of the hairpin must be least 4 bases lo
- 28: NJ algorithm is used to produc Optimal RNA 2' sturcture.
- 29: energy base algorithm type is Zuker algorithm.
- 30: Types of RNA...present in organism. (5)
- 31: The online search is for DNA and RNA is..... genbank
- 32: Rooted tree is computational expensive.
- 33: In multiple sequence algorithm doing so many progressive alignment can be slow.
- 34: DNA dictates the production of cell protein and carbohydrate

- 35: protein have long sequences ...in no?
- 36: In Exact partial matching of residue and order with room for variation in both.
- 37: DNA/RNA sequence is available on.... GENE BANK 38:
- Martine algorithm predict for structure 2 RNA.
- 39: In extraction of ORF, 1st step is obtained the 3 codons.
- 40: Example of energy based algorithm Zukar algorithm.
- 41: Dynamic programming help us to..... reduce computational cost.
- 43: When unpaired bases in 2 region binds is result form Buldges.
- 44: which one of following is not objective based method of phylogenic ... UPGMA
- 45: protein sequencing can be done by Swiss prot & UniProt.
- 46: Drugs can be better developed after understanding molecular basis of Disease
- 47: Online tools are available for homology modeling.
- 48: Insertion and deletion in alignment cause..... gaps.
- 49: Alligment of insertion and deletion can be performed by dynamic programming
- 50: Exons in proteins tend to be more conserved than intron.
- 51: Blast confined sequences from—— ID,name,spcies
- 52: ribosmes read RNA and collect amino acid from..... cell cytosol.
- 53: Simulation can predict diseases progression & outcomes.
- 54: For exact machingrequired. Traceback 55: traceback is start from high score on couple nucleotide.
- 56: Motif protein chain fold to take .
- 57: Next generation sequencing is used to known about unknown sequences of nucleotide.
- 58: RNA fold to form 2 structure.
- 59: MFOLD implement are ungraded version of Zuker.
- 60: Which has less computational cost Dynamic programming
- 61: In scoring matrices, we score
- 62: We compare all length of nucleotides in Global alignment.
- 63: we compare smaller fragment Local alignment.

- 64: Destabilized RNA molecule have (+)value.
- 65: In Zukar algorithm, we need construct all possible combination.
- 66: We use Zukar algorithm to sum up energy to find lowest matrix.
- 67: Swiss prot. is a box in uniprot.
- 68: In comparing RNA structure, we used to select which having lowest energy.
- 69: Addition of nucleotide is done by hydrogen bond
- 71: Edman degradation is used for protein sequencing.
- 72: Why do we create all possible combination of diagonals in Zukar algorithm with overall lowest energy is selected.
- 73: We can do repeated sequences alignment by slight modification in waterman model.
- 74: Length of hairpin loop is 4 bases.
- 75: "T" Threonine is an basic element....true
- 76: To align sequence what will be necessary firstly.....
- 77: Sequence of biomolecules can be known by using
- 78: Definition of Expasy: Expasy provide access to variety of online database and tools.
- 79: Structural information of a molecule can be obtained by X-ray crystallography , nuclear magnetic sequence.
- 80: Compare a protein query sequence against a nucleotide sequence database.....tBlastn
- 81: Protein formation of abnormal folding called Pseudoknots.
- 82: By using uniprot we can find sequence of.....(protein)
- 83: The 5' end of molecule capped with 7 Methyl Guanosine Triphosphate.
- 84: DNA provide blue print for building.
- 85: DNA get modified by mutation, substitution, insertion, deletion.
- 86: UPGMA is used to calculate the distance of each cluster.
- 87: Global alignment used to determine overall conversation.
- 88: if the sequence is multiple there can be several possible alignment.

- 89: can we identify the post translational modifications by .
- 90: scoring scheme has an extremely crucial role in producing optical alignment.
- 91: why is there need to predict RNA structures to know about its function
- 92: DP recombines the nucleotide recombination through a process of Traceback.
- 93: .how many open reading frame exist for each DNA sequence.6ORFs
- 94: tblastn search translated nucleotide using...Protein query
- 95: Blast can find sequencing from protein or nucleotide database it can find known and unknown sequencing.
- 96: name any two frequently used scoring matrix Blosum and pam.
- 97: From Fasta we can obtain the information on the parent organism, function and evolution history.
- 98: The correct order of progressive alignment for multiple sequencing alignment is Pair wise alignment, Guide tree, branching order in tree.
- 99: CLUSTALW can run in two modes slow/accurate, fast/appropriate.
- 100: Gene bank and UniProt have Large number of sequencing publically.
- 101: In multiple sequencing, we have Global alignment.
- 102: Ubiquitin play important role recycling protein.
- 103: 6 form of DNA ending sequencing.
- 104: Non-clustering method such as maximum parsimony may be used for making trees.
- 105: If in needleman and Smith watermark algorithm along '0' place is in.....relationship. 1st row and 1st column.
- 107: Faster active alignment by using local alignment and protein nucleotide. pairwise alignment
- 109: Expasy was develop by Swiss Bioinformatics Institute (SIB)
- 110: .In phylogenetics tree , ancestor are placed at Root/bottom/ancestor node trees.
- 111: .In UPQMA two sequence are .
- 112: .When unpaired bases in 2 region binds itself form Buldges.

113: Zukers algorithm is use to predict computational and destabilizing energy?

114: In Nussinov Jacob algorithm which turn is added to compare to score for diagonal position set diagonal

and lower tri diagonal to zero.

115: Trackback begins from the position in Nussinov Jacobson score matrix. 4rth possition

1) Exact matching require which type of nucleotide-same number of nucleotide.

2) Accurate solution of protein structure is one of the toughest problems 3)

Tract back is overlap matches can start from any position in scoring matrix.

5) The matrix hasscores. Positive and negative.

6) MASCOT....._can search people mass finger printing and shotgun proteomics data

set 7) Spectrometer measures the protein by Mass/charge

8) Simulation of the folding process depends Energy function

9) Information about DNA/RNA is available on gene bank

1.Domain are----

2.RNA brick database-----

3.overcome homology modelling limitations by-----

4.homology predicts structure of protein-----

5.aromatic amino acids-----

- 6.DNA/RNA sequence----database

7.mRNA does not have-----

- 8. BLAST can find sequences-----

- 9.MS based proteomics begins measurement of -----

10. alpha carbon atom can be obtain by---

11. Ab Initio methods can predict the sytructure on the bases of _____, charg volume mass.

12. The value of one angstrom _____.

13. _____ - amino acid are required for alpha helices.

14. Blosum aberration_____.

15. The pka value of valin_____.

16. The sequences that are similar we _____ modeling.

17. Exact matching require which type of nucleotide_____ -
18. _____ - coordinates the alpha carbon.
19. We start from ___top right___ in global alignment.
19. In structure visualization the sequence are visualized as cartoon_____.
20. pK value of Aspartic acid 3.9
21. ExPASy was made by _____ -.
22. Dot plots employ dot matrix with two sequence plotted represents.....pairwise
23. Which give info about precursor protein..... PSTS
24. mgf files format can be converted to
25. Molecular evolution can be caused byinsertion deletion substitution.
26. XYZ position of position is available in _____
27. We can predict better by _____.
28. The amino acid join together by _____ bond.
29. One and 4 amino in alpha helix joined by _____ bond.
30. Cn3d is an online tool used for_____.
31. Dot plots employ dot matrix with two sequences plotted on_____ -
40. Fasta Was Developed In(Ans1988)
41. In Corey Pauling Model Of The Protein Each Atom Is Represented By(Solid Sphere)
42. How Many Storage Use For Structural Modeling (3)
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48. RNA Don't Have Structure???Quaternary
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50. Coding Region Of DNA
51. Blast Use For???Unknown And Known Protein

52. Expasy Develop

A. Bond Form Between C And N Terminus??? Hydrogen Bond

B. X.Y.Z Position Of Protein Are Available By??? Pdb

C. Technique Use To Measure The Mass Of Protein??? Mass Spectrometry

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E. Positive Charge Amino Acid??? Lysine And Arginine

F. Mrna Has??? Single Strand

G. C.Alpha Atom Are Traced To Reconstruct??? 3d Structure Of Protein
Bond

H. Form Between Two Atoms??? Energy Release

I. Provide Clue Of Precursor Protein Peptide Sequence??? Pst

J. Amino Acid Required For Beta Sheet??? 3-5

K. Are Formed By Bonding Between C And N In Protein Backbone??? Alpha Helix

L. Blast Stand For??? Basic Local Alignment Search Tool

Amino Acid

M. Have Unique Trait Which Can Help In??? Protein Folding Align

N. Comparison Between Dna/Dna And Dna/Protein??? Fasta.

O. Ratio Use To Calculate Mass Of Protein??? M/Q.

P. Treading Not Required For??? Structure Of Entire Backbone.

Q. Needle Man Wound Man Or Smith Water Man Algorithm Use For??? Local And
Global Alignment.

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